ABRAR SAMIN

Lecturer, Dept. of Software Engineering, Daffodil International University · AI/ML Researcher

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PROFESSIONAL SUMMARY

Lecturer in the Department of Software Engineering at Daffodil International University, and a Computer Science and Engineering graduate of BRAC University (CGPA 3.97/4.00), graduating among the top of the department, with four IEEE-published research papers in machine learning, computer vision, and explainable AI, and three further papers under review in 3D computer vision. Served as an Undergraduate Teaching Assistant for both Data Structures and Algorithms, mentoring 200+ students, and contributed to BRAC University-funded research under the Research Seed Grant Initiative (RSGI), including AI-based assistive technology for accessibility. Strong foundation in machine learning, NLP, computer vision, and full-stack development, with a keen interest in teaching, mentoring, and advancing applied AI research in a university setting.

RESEARCH INTERESTS

Machine Learning · Computer Vision · 3D Vision & SLAM · Explainable AI (XAI) · Trustworthy AI · Medical Imaging · Biomedical Signal Processing · Embedded / Edge AI · Assistive Technology

ACADEMIC APPOINTMENTS

Lecturer **Sep. 2026 – Present**

Department of Software Engineering, Daffodil International University — Dhaka, Bangladesh

- Teaching three undergraduate courses with their accompanying laboratory sections: Software Quality Assurance Testing (SE312), Data Structure (SE131), and Algorithm Analysis (SE214).
- Continuing research in machine learning, computer vision, and explainable AI alongside teaching.

EDUCATION

BRAC University **2022 – 2026**

B.Sc. in Computer Science and Engineering — Dhaka, Bangladesh

CGPA: 3.97 / 4.00 — graduated among the top of the department

Thesis: *CAMDet: Camera-Agnostic Monocular Metric 3D Object Detection with Learned Intrinsic Feature Conditioning* (under review)

PUBLICATIONS

JOURNAL ARTICLES

[J1] Reshad Ul Karim, Sammam Mahdi, **Abrar Samin**, Aniqua Nusrat Zereen, M. Abdullah-Al-Wadud, and Jia Uddin, "Optimizing Stroke Recognition with MediaPipe and Machine Learning: An Explainable AI Approach for Facial Landmark Analysis," *IEEE Access*, vol. 13, pp. 32636–32660, 2025, doi: 10.1109/ACCESS.2025.3550577.

CONFERENCE PAPERS

[C1] Tasnim Ferdous, Reshad Ul Karim, **Abrar Samin**, Sammam Mahdi, Himika Tasnim, and Aniqua Nusrat Zereen, "Machine Learning Approaches in Photoplethysmography-Based Sleep Stage Classification," *in Proc. 2024 IEEE 2nd Int. Conf. on Electrical, Automation and Computer Engineering (ICEACE)*, Changchun, China, Dec. 2024, pp. 123–128, doi: 10.1109/ICEACE63551.2024.10898858.

[C2] Tasnim Ferdous, Reshad Ul Karim, **Abrar Samin**, Sammam Mahdi, and Aniqua Nusrat Zereen, "Improved Photoplethysmography-Based Four-Stage Sleep Classification with Explainable AI-Driven Machine Learning," *in Proc. 2024 IEEE 2nd Int. Conf. on Electrical, Automation and Computer Engineering (ICEACE)*, Changchun, China, Dec. 2024, pp. 117–122, doi: 10.1109/ICEACE63551.2024.10898853.

[C3] Sammam Mahdi, Reshad Ul Karim, Tareq Sujat, Syeda Maliha Tabassum, **Abrar Samin**, and Aniqua Nusrat Zereen, "Vision-based Hand Gesture Virtual Keyboard-Mouse framework with Bilingual Next-word prediction," *in Proc. 2026 23rd Int. Joint Conf. on Computer Science and Software Engineering (JCSSE)*, Bangkok, Thailand, Jun. 2026, pp. 638–643, doi: 10.1109/JCSSE68839.2026.11596610.

ONGOING RESEARCH PROJECTS

UNDER REVIEW & PREPRINTS

CAMDet: Camera-Agnostic Monocular Metric 3D Object Detection with Learned Intrinsic Feature Conditioning **Under review**

*Reshad Ul Karim, Sammam Mahdi, **Abrar Samin**, Syeda Maliha Tabassum, Rakin Bin Rabbani, and Aniqua Nusrat Zereen — B.Sc. thesis*

A single DINOv2-DETR model with FiLM intrinsic conditioning predicts 2D boxes, per-object metric depth, and amodal 3D boxes that transfer to unseen cameras: 46.9 AP3D at IoU 0.25 on SUN RGB-D, +6.3 AP3D over a virtual-depth baseline on held-out sensors, and 166 FPS (ViT-S), with one checkpoint deploying unmodified from desktop GPU to an embedded Jetson module.

CADENCE: Sensor-Grounded World-Consistent Gaussian SLAM for Collaborative Mobile Reconstruction — **Under review**

3D Vision · Collaborative SLAM — under double-blind review; arXiv preprint forthcoming

Collaborative 3D Gaussian-splatting SLAM across smartphones in which only deterministic geometric and physical tests can accept a cross-device link and per-modality confidences gate what a peer may change, so a degraded device cannot corrupt the shared map; false links stay below 0.5% on eight held-out Matterport3D houses, and a native iPhone app maps 15 Hz LiDAR in real time (20.3 ms p95 pose time).

Less Than Meets the Eye: Action-Set Identifiability and Principled Abstention for 3D Physical Geometry — **Under review**

3D Vision · Identifiability — under double-blind review; arXiv preprint forthcoming

Intervene3D decides whether a bounded set of camera motions can tell a real scene apart from its mirror reflection or an on-screen image, abstaining when no motion separates the explanations; abstention lowers false physical certainty from 0.832 to 0.000 on a pixel-identical benchmark, and viewpoint selection lowers ambiguous-region error by 31.3% on real photographs.

MANUSCRIPTS READY FOR SUBMISSION

On-Device Sleep Staging from Wrist Actigraphy and Heart-Rate Variability: A CNN-SE-BiLSTM Sequence Modeling Approach — **Manuscript complete**

***Abrar Samin**, Himika Tasnim, and Aniqua Nusrat Zereen — preparing for journal submission*

Leakage-controlled, subject-independent four-stage sleep staging from a wrist wearable: 77.2% accuracy and Cohen's κ of 0.68 (+7.9 points over logistic regression), a 3.28× distilled model for on-device use, and total sleep time that agrees with polysomnography (r = 0.96).

Patient-Referenced Evaluation of Clinical Findings and Re-Identification in Report-Conditioned Synthetic Chest Radiographs — **Manuscript complete**

Medical Imaging · Generative AI — preparing for journal submission

A patient-referenced protocol for what diffusion-generated chest X-rays carry beyond their prompt: across three generators and three cohorts, renders show findings the prompt omits far less often than the patient's own radiograph, and re-identification against prompt-matched distractors falls close to chance.

IN PROGRESS

Hybrid Deep Learning and Signal–Spectrogram Fusion for EEG Classification — **In progress**

Biomedical Signals · EEG — code and experiments complete; manuscript in preparation

A CNN–BiLSTM–GRU hybrid benchmarked against a Random Forest baseline with a full architecture ablation, and a dual-branch network fusing the raw signal with its STFT spectrogram via cross-attention, with gradient-based saliency for interpretability.

RESEARCH EXPERIENCE

Undergraduate Researcher — **2024 – Present**

Research Seed Grant Initiative (RSGI), BRAC University — Dhaka, Bangladesh

- Contributed to a BRAC University-funded research project under the Research Seed Grant Initiative (RSGI) developing an AI-powered assistive navigation system to support safe, independent mobility for blind and low-vision users.
- Worked on computer vision and on-device (edge) AI components for a low-cost wearable platform, advancing the system from a validated proof-of-concept toward a deployable prototype intended for real-world, ethics-approved field trials with blind and low-vision users.

TEACHING EXPERIENCE

Undergraduate Teaching Assistant — **2024 – Feb. 2026**

Department of Computer Science and Engineering, BRAC University — Dhaka, Bangladesh

- Served as Teaching Assistant for both Data Structures and Algorithms, supporting lectures, lab sessions, and consultation hours — reaching 200+ students.
- Guided students through programming assignments and algorithmic problem-solving, strengthening their understanding of fundamental data structures and algorithm design.
- Graded assignments and exam scripts and gave constructive, individualized feedback to help students improve their performance.

SELECTED PROJECTS

WeHeal — Healthcare Innovation Platform — **weheal.live**

An AI-powered, all-in-one healthcare platform for patient care and medical diagnostics — online doctor consultations, appointments, prescriptions, and emergency support in one place. Built with a team and awarded 1st Runner-Up at BRAC University's System Analysis & Design Project Showcase (2025).

Autonomous ArUco-Marker Multi-Terrain Service Bot — **Video demo**

An autonomous service robot that navigates varied terrain using ArUco fiducial-marker detection for localization and task guidance, pairing computer vision with real-time control on a mobile robotic platform.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript

Frameworks & Libraries: Node.js, Express.js, MediaPipe

Databases: MySQL, MongoDB

Tools & Technologies: Git, REST APIs, Machine Learning frameworks, Computer Vision libraries

HONORS & AWARDS

- Dean's List — recognized for academic excellence, BRAC University.
- 2nd Place — BRAC University RoboSoccer Competition.
- 1st Runner-Up — System Analysis and Design Project Showcase, Summer 2025 (for WeHeal).

LANGUAGES

English — Professional working proficiency · Bangla — Native proficiency

REFERENCES

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